

# A STOCHASTIC VOLATILITY MODEL PARAMETER CALIBRATION ROUTINE

ABSTRACT. We specify the routine that calibrates the five Heston (1993) parameters  $\Phi = (\theta, \kappa, \eta, \rho, v_0)$  to a day of CBOE S&P 500 (SPX) intraday option trades. The routine performs *one* calibration per trading day over a pooled, moneyness-normalised implied-volatility surface, and is framed throughout as a box-constrained nonlinear least-squares problem solved from multiple seeds. We make explicit every modelling choice: the volume-weighted reference spot  $S_{\text{ref}}$ , the moneyness-normalised and grid-snapped strike  $K^*$ , the surface-coverage requirements, the in-engine relative-price objective, the implied-volatility (“IV-space”) selection and acceptance gate, and the boundary-pegging rejection rule.

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## 1. The model

Under the risk-neutral measure, with constant risk-free rate  $r$  and continuous dividend yield  $g$ , the log-price  $X_t = \ln S_t$  and instantaneous variance  $v_t$  of the SPX follow the Heston (1993) dynamics

$$\begin{aligned} (1) \quad dX_t &= (r - g - \tfrac{1}{2}v_t) dt + \sqrt{v_t} (\rho dW_t + \sqrt{1 - \rho^2} dB_t), \\ (2) \quad dv_t &= \kappa(\theta - v_t) dt + \eta\sqrt{v_t} dW_t, \end{aligned}$$

where  $W, B$  are independent Brownian motions, so  $\text{corr}(dX_t, dv_t) = \rho dt$ . The parameter vector is

$$(3) \quad \Phi = (\theta, \kappa, \eta, \rho, v_0) \in \mathbb{R}^5,$$

with  $\theta > 0$  the long-run variance,  $\kappa > 0$  the mean-reversion speed,  $\eta > 0$  the volatility of variance (QuantLib's `sigma`),  $\rho \in [-1, 1]$  the leverage correlation, and  $v_0 > 0$  the initial variance. The Feller quantity (Feller, 1951)

$$(4) \quad \mathcal{F}(\Phi) = 2\kappa\theta - \eta^2$$

is reported per fit;  $\mathcal{F}(\Phi) \geq 0$  guarantees  $v_t > 0$  almost surely. It is not imposed as a constraint (short-tenor SPX fits routinely violate it without being otherwise unreasonable; see §6).

**1.1. Pricing operators.** Fix a valuation with maturity  $\tau > 0$  (in years), strike  $K$ , and type  $w \in \{+1, -1\}$  (+1 call, -1 put). Write the forward and discount factor under flat curves as

$$(5) \quad F(S, \tau) = S e^{(r-g)\tau}, \quad D(\tau) = e^{-r\tau}.$$

**Black price** (Black, 1976; Black and Scholes, 1973). With total standard deviation  $\Sigma = \sigma\sqrt{\tau}$ ,

$$\begin{aligned} (6) \quad C_B(\sigma; S, K, \tau, w) &= w D(\tau) \left[ F \mathcal{N}(wd_+) - K \mathcal{N}(wd_-) \right], \\ d_{\pm} &= \frac{\ln(F/K)}{\Sigma} \pm \tfrac{1}{2}\Sigma, \end{aligned}$$

where  $\mathcal{N}$  is the standard normal CDF and  $F = F(S, \tau)$ .

**Heston price.** The Heston model price  $C_H(\Phi; S, K, \tau, w)$  is the semi-analytic value

$$\begin{aligned} (7) \quad C_H(\Phi; S, K, \tau, +1) &= S e^{-g\tau} P_1 - K e^{-r\tau} P_2, \\ C_H(\cdot, -1) &= C_H(\cdot, +1) - S e^{-g\tau} + K e^{-r\tau}, \end{aligned}$$

with the in-the-money probabilities, for  $j \in \{1, 2\}$ ,

$$(8) \quad P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[ \frac{e^{-iu \ln K} f_j(u; \Phi, \tau, \ln F)}{iu} \right] du,$$

and characteristic functions  $f_j = \exp(C_j + D_j v_0 + iu \ln F)$  given by the standard coefficients

$$\begin{aligned} (9) \quad C_j &= \frac{\kappa\theta}{\eta^2} \left[ (b_j - \rho\eta iu + d_j)\tau - 2 \ln \frac{1 - \gamma_j e^{d_j\tau}}{1 - \gamma_j} \right], \\ D_j &= \frac{b_j - \rho\eta iu + d_j}{\eta^2} \frac{1 - e^{d_j\tau}}{1 - \gamma_j e^{d_j\tau}}, \end{aligned}$$

$$(10) \quad d_j = \sqrt{(\rho\eta iu - b_j)^2 - \eta^2(2\mu_j iu - u^2)}, \quad \gamma_j = \frac{b_j - \rho\eta iu + d_j}{b_j - \rho\eta iu - d_j},$$

where  $\mu_1 = \frac{1}{2}$ ,  $\mu_2 = -\frac{1}{2}$ ,  $b_1 = \kappa - \rho\eta$ ,  $b_2 = \kappa$ . In code this integral is evaluated by the QuantLib (QuantLib) `AnalyticHestonEngine` (which uses a branch-stable formulation of (8)); equations (7)–(8) define the operator it realises.

**Implied volatility.** The Black-implied volatility of a price  $C$  is the inverse of (6) in its first argument,

$$(11) \quad \mathcal{I}(C; S, K, \tau, w) = \{ \sigma > 0 : C_B(\sigma; S, K, \tau, w) = C \},$$

computed numerically (QuantLib `blackFormulaImpliedStdDev` on the forward (5), returning  $\Sigma$ , then  $\sigma = \Sigma/\sqrt{\tau}$ ). This operator defines the IV-space residuals of §5.

## 2. Data and per-day notation

The pipeline ingests CBOE trade-level files, one per trading day  $d$ , restricted upstream to out-of-the-money (OTM) quotes. A day is a set of trade records indexed by  $i$ , each carrying

$$(12) \quad (t_i, K_i, w_i, q_i, P_i, \sigma_i, S_i, \tau_i),$$

being respectively the quote timestamp, strike, type  $w_i \in \{+1, -1\}$ , traded size, trade price, trade-implied volatility, the underlying level *at trade time*  $t_i$ , and the days to maturity. Maturities enter the engine as  $\tau_i$  in years under Actual/365, i.e.  $T_i = \tau_i/365$ .

The day is rate-stamped with  $r = r(d)$  and  $g = g(d)$ , the most recent risk-free and dividend rates on or before  $d$  (a flat-forward curve for each; under flat curves the evaluation date is immaterial to  $T_i$ , so it is set to the QuantLib “today”).

**Pre-filter.** We keep only economically usable, Heston-appropriate quotes:

$$(13) \quad \mathcal{D} = \{ i : \sigma_i > 0 \text{ and } \tau_i \geq \tau_{\min} \}.$$

The floor  $\tau_{\min}$  (Table 1) drops ultra-short maturities, which the diffusive Heston smile fits poorly and which otherwise drive  $\eta, \kappa$  to Feller-violating extremes that pollute the pooled fit.

## 3. Handling intraday spot movement

A Heston calibration is anchored to a *single* spot  $S$ , yet the trades in  $\mathcal{D}$  occur across an intraday range of underlying levels  $\{S_i\}$ . We reconcile the two with one reference spot and a moneyless re-striking.

**Reference spot.** Define the volume-weighted reference spot

$$(14) \quad S_{\text{ref}} = \frac{\sum_{i \in \mathcal{D}} q_i S_i}{\sum_{i \in \mathcal{D}} q_i}.$$

The intraday range and its flag are

$$(15) \quad R = \frac{\max_i S_i}{\min_i S_i} - 1, \quad \text{HIGH\_MOVE} = \mathbf{1}\{R > R_{\max}\}.$$

A flagged day ( $R > R_{\max}$ ) is still calibrated and written, but the normalisation below is recorded as strained (its  $S_{\text{ref}}$  should be treated with suspicion).

**Moneyness normalisation and grid snap.** Each trade retains its moneyness at trade time and is re-struck to the reference spot, then snapped to the SPX strike grid of width  $\Delta$ :

$$(16) \quad m_i = \frac{K_i}{S_i}, \quad K_i^* = \Delta \cdot \text{round}\left(\frac{m_i S_{\text{ref}}}{\Delta}\right).$$

The rationale is sticky-moneyness: over a session the implied-volatility smile is approximately stationary in  $K/S$  rather than in the absolute strike. Re-striking by (16) therefore re-centres the whole day onto  $S_{\text{ref}}$  so that trades executed at different  $S_i$  but the *same* moneyness share a common surface column. The grid snap collapses the continuum of  $K_i^*$  values (one per trade) back onto a clean strike lattice. Pricing diagnostics (§7) nonetheless use each contract's *original*  $(S_i, K_i)$ , since the calibrated  $\Phi$  is spot-independent.

#### 4. Surface construction

From the normalised trades we build a partial  $K^* \times \tau$  implied-volatility surface, selecting the most liquid, nearest-the-money quotes.

**Maturity selection.** For each distinct maturity  $\tau$  present in  $\mathcal{D}$ , let  $V(\tau) = \sum_{i \in \mathcal{D}: \tau_i = \tau} q_i$  be its traded volume. Keep the  $N_t$  highest-volume maturities,

$$(17) \quad \mathcal{T} = \text{top-}N_t \text{ maturities by } V(\tau).$$

**Strike selection (per maturity, per wing).** Within each kept maturity  $\tau \in \mathcal{T}$ , separate calls ( $w_i = +1$ , OTM with  $K_i^* \gtrsim S_{\text{ref}}$ ) from puts ( $w_i = -1$ , OTM with  $K_i^* \lesssim S_{\text{ref}}$ ) and keep the  $n_k$  nearest-the-money distinct strikes on each wing:

$$(18) \quad \mathcal{K}_\tau = \underbrace{\{n_k \text{ largest } K^* \text{ among puts at } \tau\}}_{\text{put wing}} \cup \underbrace{\{n_k \text{ smallest } K^* \text{ among calls at } \tau\}}_{\text{call wing}}.$$

**Pivot to a surface.** For each retained cell  $(k, \tau)$  with  $k \in \mathcal{K}_\tau$ , set the surface entry to the implied volatility of the *latest* trade in that cell,

$$(19) \quad \hat{\sigma}_{k,\tau} = \sigma_{i^*}, \quad i^* = \arg \max_{i: K_i^* = k, \tau_i = \tau} q_i,$$

and leave it missing otherwise. Let

$$(20) \quad \mathcal{N} = \{(k, \tau) : \hat{\sigma}_{k,\tau} \text{ defined}\}$$

be the set of populated cells; these become the calibration targets (helpers).

**Coverage requirement.** To keep the five-parameter fit identified we require a genuinely two-dimensional surface:

$$(21) \quad |\{\tau : (\cdot, \tau) \in \mathcal{N}\}| \geq M_t, \quad |\{k : (k, \cdot) \in \mathcal{N}\}| \geq M_k, \quad |\mathcal{N}| \geq M_{\text{cell}}.$$

A day failing (21) (or with no usable maturities) is dropped: it contributes no calibration and clears its stale diagnostics.

### 5. The optimization problem

Let  $\mathcal{N}$  index the targets with market data  $\{(\hat{\sigma}_n, K_n^*, T_n, w_n)\}_{n \in \mathcal{N}}$ , all priced at the single spot  $S_{\text{ref}}$ . Define the feasible box

$$(22) \quad \mathcal{B} = \prod_{j=1}^5 [\ell_j, h_j] \subset \mathbb{R}^5, \quad (\ell, h) \text{ in the } (\theta, \kappa, \eta, \rho, v_0) \text{ order of Table 2.}$$

**5.1. Residuals.** Two residual systems are used: a price-space system that the local optimizer minimises, and an IV-space system that selects among restarts and gates acceptance. The in-engine objective is configurable. It defaults to the price-space system below, and can instead be set to the IV-space system of (25). The choice changes only what each restart minimises. Selection (30) and the acceptance gate (32) always run on the IV-space error  $J_{\text{iv}}$ , so they are unaffected by it.

**Price-space (in-engine objective).** For each target the market value is the Black price at the quoted vol,  $C_n^{\text{mkt}} = C_{\text{B}}(\hat{\sigma}_n; S_{\text{ref}}, K_n^*, T_n, w_n)$ , and the relative-price residual is

$$(23) \quad \varepsilon_n^{\text{px}}(\Phi) = \frac{C_{\text{H}}(\Phi; S_{\text{ref}}, K_n^*, T_n, w_n) - C_n^{\text{mkt}}}{C_n^{\text{mkt}}}.$$

The associated objective is the root-mean-square

$$(24) \quad J_{\text{px}}(\Phi) = \left( \frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{px}}(\Phi)^2 \right)^{1/2}.$$

The per-target residual  $\varepsilon_n^{\text{px}}$  is the (relative-price) error each QuantLib helper exposes. Levenberg–Marquardt (Marquardt, 1963) is a sum-of-squares method, so it drives the squared residual norm  $\sum_n (\varepsilon_n^{\text{px}})^2 = |\mathcal{N}| J_{\text{px}}(\Phi)^2$  to a local minimum (equivalently  $J_{\text{px}}$ , with which it shares a minimiser).

**IV-space (selection and gate).** Invert each Heston price back to a Black vol via (11) and compare to the market vol:

$$(25) \quad \begin{aligned} \sigma_n^{\text{mod}}(\Phi) &= \mathcal{I}(C_{\text{H}}(\Phi; S_{\text{ref}}, K_n^*, T_n, w_n); S_{\text{ref}}, K_n^*, T_n, w_n), \\ \varepsilon_n^{\text{iv}}(\Phi) &= \sigma_n^{\text{mod}}(\Phi) - \hat{\sigma}_n, \end{aligned}$$

with objective

$$(26) \quad J_{\text{iv}}(\Phi) = \left( \frac{1}{|\mathcal{N}'|} \sum_{n \in \mathcal{N}'} \varepsilon_n^{\text{iv}}(\Phi)^2 \right)^{1/2}, \quad \mathcal{N}' = \{n : \sigma_n^{\text{mod}}(\Phi) \text{ inverts finitely}\}.$$

$J_{\text{iv}}$  is measured in volatility points; it is the natural surface-fit metric and, unlike  $J_{\text{px}}$ , is not inflated by cheap deep-OTM wings (a unit price error on a near-worthless contract is a large *relative* price error but a small vol error).

**5.2. Multi-start local optimization.** A single start into a non-convex problem is fragile (Mikhailov and Nögel, 2003), so we restart from a small, data-seeded set. Let the surface variance level be

$$(27) \quad \hat{v} = \text{clip}\left(\left(\text{median}_{(k,\tau) \in \mathcal{N}} \hat{\sigma}_{k,\tau}\right)^2, [10^{-3}, 0.25]\right),$$

and define the seed set  $\mathcal{S} = \{s_1, \dots, s_6\} \subset \mathcal{B}$  of  $(v_0, \kappa, \theta, \eta, \rho)$  tuples

$$(28) \quad \mathcal{S} = \begin{cases} (\hat{v}, 1.0, \hat{v}, 0.50, -0.70), & (\hat{v}, 3.0, \hat{v}, 1.00, -0.50), \\ (\hat{v}, 0.5, \hat{v}, 0.30, -0.90), & (\hat{v}, 5.0, \hat{v}, 0.80, -0.60), \\ (0.8\hat{v}, 2.0, 1.2\hat{v}, 0.60, -0.75), & (\hat{v}, 8.0, \hat{v}, 1.20, -0.40). \end{cases}$$

For each seed  $s \in \mathcal{S}$  we solve the box-constrained nonlinear least-squares problem

$$(29) \quad \hat{\Phi}_s = \arg \min_{\Phi \in \mathcal{B}} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{px}}(\Phi)^2 \quad (\text{LM, started at } \Phi^{(0)} = s),$$

using Levenberg–Marquardt under the box constraint  $\mathcal{B}$  (`QuantLib NonhomogeneousBoundaryConstraint`). The fit retained for the day is the restart with the smallest *IV-space* error,

$$(30) \quad \Phi^* = \arg \min_{s \in \mathcal{S}} J_{\text{iv}}(\hat{\Phi}_s).$$

Ranking by  $J_{\text{iv}}$  (not  $J_{\text{px}}$ ) removes the dependence on any single arbitrary guess and selects the economically best surface fit.

## 6. Acceptance gate

$\Phi^*$  is accepted only if it fits the surface tightly *and* lies in the interior of the box. Let the boundary-pegging predicate, with tolerance fraction  $\delta$ , be

$$(31) \quad \text{PEGGED}(\Phi) = \bigvee_{j=1}^5 \left[ (\Phi_j - \ell_j) < \delta(h_j - \ell_j) \quad \vee \quad (h_j - \Phi_j) < \delta(h_j - \ell_j) \right].$$

The day is accepted iff

$$(32) \quad \boxed{J_{\text{iv}}(\Phi^*) \leq \varepsilon_{\text{iv}} \quad \text{and} \quad \neg \text{PEGGED}(\Phi^*)}.$$

The first clause is the IV-space gate ( $\varepsilon_{\text{iv}} \approx 2$  vol points). The second rejects boundary fits: a parameter resting on a bound is the optimizer hitting a wall, not an identified value, so it is treated as a non-fit regardless of  $J_{\text{iv}}$ . The Feller quantity (4) is reported but never gates, since short-tenor SPX fits can violate it while remaining otherwise sound. A rejected or too-thin day yields no parameter row.

**Why an IV-space gate.** An earlier version gated on  $J_{\text{px}}$  directly. On a pooled day rich in deep-OTM wings,  $J_{\text{px}}$  is dominated by those cheap contracts and rejects fits that are excellent in vol points. Replacing the gate metric with  $J_{\text{iv}}$  aligns the acceptance criterion with the units the surface is quoted in; under it the genuine pooled-surface fit is  $\approx 0.8$ – $1.4$  vol points, and the residual rejections are boundary-pegged  $\kappa$  or  $\rho$  (a separate identification lever), not fit-quality failures.

## 7. Output and diagnostics

An accepted day yields one parameter row keyed by date,

$$(33) \quad (d, S_{\text{ref}}, r, g, \theta^*, \kappa^*, \rho^*, \eta^*, v_0^*, \mathcal{F}(\Phi^*), J_{\text{iv}}, J_{\text{px}}, \dots, R, \text{HIGH\_MOVE}),$$

together with coverage counts ( $|\mathcal{T}|, \#\text{strikes}, |\mathcal{N}|$ ) and total volume. Because the calibration is once-daily, these rows accumulate into a single file, regenerated each run from the accepted days. For repricing diagnostics each retained contract is valued under  $\Phi^*$  at its *original*  $(S_i, K_i)$ , not  $(S_{\text{ref}}, K^*)$ , since the structural

parameters are spot-independent; this gives an honest comparison against the actual trade price  $P_i$ .

## 8. Constants

| Symbol                    | Code name      | Value     | Role                                    |
|---------------------------|----------------|-----------|-----------------------------------------|
| $\tau_{\min}$             | MIN_DTM        | 7 days    | minimum maturity (drop ultra-short)     |
| $R_{\max}$                | MAX_MOVE_PCT   | 0.03      | intraday-range flag threshold           |
| $\Delta$                  | STRIKE_GRID    | 5.0       | SPX strike lattice for $K^*$ snap       |
| $N_t$                     | MAX_NT         | 12        | maturities kept (by volume)             |
| $n_k$                     | MAX_NK         | 8         | strikes kept per wing                   |
| $M_t$                     | MIN_MATS       | 3         | minimum maturities in surface           |
| $M_k$                     | MIN_STRIKES    | 5         | minimum strikes in surface              |
| $M_{\text{cell}}$         | MIN_CELLS      | 12        | minimum populated cells $ \mathcal{N} $ |
| $\varepsilon_{\text{iv}}$ | IV_RMSE_ACCEPT | 0.02      | IV-space acceptance gate (vol pts)      |
| $\delta$                  | BOUND_TOL      | $10^{-3}$ | boundary-pegging tolerance fraction     |

TABLE 1. Surface, selection, and gate constants.

|              | $\theta$  | $\kappa$  | $\eta$    | $\rho$   | $v_0$     |
|--------------|-----------|-----------|-----------|----------|-----------|
| lower $\ell$ | $10^{-4}$ | $10^{-2}$ | $10^{-2}$ | $-0.999$ | $10^{-4}$ |
| upper $h$    | 1.0       | 20.0      | 2.0       | 0.5      | 1.0       |

TABLE 2. Box bounds  $\mathcal{B}$ , in the `model.params()` order  $(\theta, \kappa, \eta, \rho, v_0)$ . The  $\rho$  upper bound is kept slightly positive (equity leverage implies  $\rho < 0$ ) and  $\eta$  is capped at 2.0 (SPX vol-of-vol  $\sim 0.3$ – $1.2$ ).

## 9. Empirical findings

We ran the routine over the full CBOE SPX trade history available to us, spanning 3,215 trading days from 3 January 2012 to 9 October 2024. Of these, 1,742 days (54.2%) passed the acceptance gate (32) and contribute a parameter row; the 1,473 rejected days are logged with their cause. The rejection split is dominated by boundary pegging: 1,369 days (93% of rejections) are rejected for a pegged parameter, 103 for an IV-space miss ( $J_{\text{iv}} > \varepsilon_{\text{iv}}$ ), and 1 for having no usable trades. Pegging (almost always  $\kappa \rightarrow 20$  or  $\rho \rightarrow -0.999$  hitting the box wall) is therefore the binding constraint on the accept rate, not surface quality.

**Fit quality.** On the accepted set the surface fit is excellent. The IV-space error (26) has a median of 0.48 vol points (mean 0.55, 90th percentile 0.90, maximum 1.95), so  $J_{\text{iv}}$  sits far inside the 2-vol-point gate on essentially every accepted day and never binds in practice. Acceptance is governed by the no-pegging clause alone. An independent repricing check (each contract valued under  $\Phi^*$  at its *original*  $(S_i, K_i)$  and compared to the trade price) confirms this: the per-day median relative price error is 6.6% and its 90th percentile 10.5%, figures inflated by the deep-OTM wings whose near-zero prices make any absolute miss a large relative one. This is exactly the effect that motivated gating in IV space rather than price space (§6). A separate validator clears 1,737 of 1,742 accepted days (99.7%); the 5 hard failures are repricing-fit outliers, not parameter pathologies.

**Parameter levels.** The accepted parameters are economically sensible and stable across the twelve-year sample. The leverage correlation is consistently and strongly negative, median  $\rho = -0.78$  (10th–90th percentile  $[-0.89, -0.61]$ ), as equity-index smiles require. Long-run variance median  $\theta = 0.053$  ( $\approx 23\%$  annualised vol) and initial variance median  $v_0 = 0.016$  ( $\approx 13\%$ ) bracket realistic SPX levels, with  $v_0$  the lower of the two on the typical low-realised-vol day. Mean reversion median  $\kappa = 2.19$  and vol-of-vol median  $\eta = 0.88$  are likewise in the usual SPX range. Because the gate rejects any boundary-pegged fit, the accepted set is pegging-free by construction: only 15 days carry  $\rho$  below  $-0.99$  and none reach the  $\kappa$  ceiling. A clean interior set is what survives the gate, not evidence that pegging is solved on the underlying population.

**The Feller condition is the standout caveat.** The gate does not test Feller, and short-tenor SPX fits violate it routinely:  $\mathcal{F}(\Phi^*) = 2\kappa\theta - \eta^2 < 0$  on 1,729 of 1,742 accepted days (99.3%), with median  $\mathcal{F} = -0.53$ . The driver of the violation is a large vol-of-vol relative to  $\kappa\theta$ ;  $\eta$  exceeds 1.5 on 161 days (9.2%) and approaches its 2.0 cap (max 1.996). A pinned  $v_0$  that disagrees with the at-the-money level appears on only 4 days and a degenerate  $\kappa$  on 7, so among the validator’s soft flags Feller violation and an elevated  $\eta$  are by far the most common. Together they mark these fits as ones whose variance path can reach zero, even where the surface fit itself is tight.

## 10. Example option prices

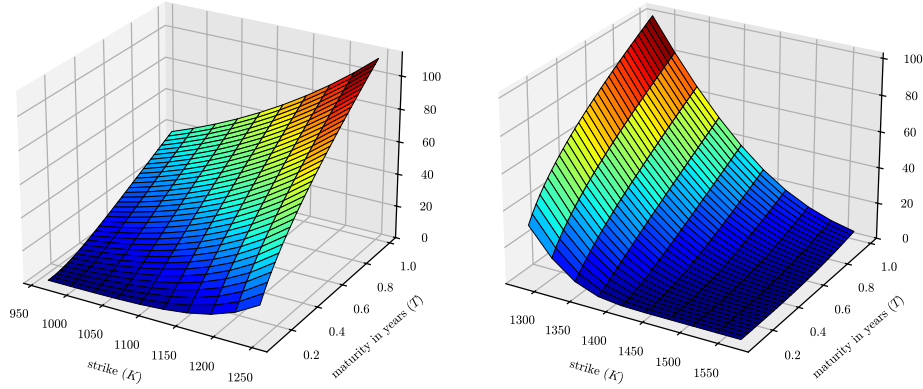


FIGURE 1. Heston OTM option prices: puts wing (left) and calls wing (right).



## 11. Algorithm

Algorithm 1 states the per-day routine; the driver applies it to every trading day and accumulates accepted rows into a single file (§7).

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### Algorithm 1 Per-day Heston calibration

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**Require:** day  $d$  of OTM trades; rates  $r(d), g(d)$ ; constants of Table 1

- 1:  $\mathcal{D} \leftarrow \{i : \sigma_i > 0, \tau_i \geq \tau_{\min}\}$  (pre-filter (13))
- 2: **if**  $r(d)$  or  $g(d)$  undefined on/before  $d$  **then**
- 3:   **return**  $\emptyset$  (no rate)
- 4: **end if**
- 5:  $S_{\text{ref}} \leftarrow (\sum q_i S_i) / (\sum q_i)$ ;  
 $R \leftarrow \max_i S_i / \min_i S_i - 1$ ;  
 $\text{HIGH\_MOVE} \leftarrow \mathbf{1}\{R > R_{\max}\}$
- 6:  $K_i^* \leftarrow \Delta \text{round}(m_i S_{\text{ref}} / \Delta)$  with  $m_i = K_i / S_i$  (re-strike (16))
- 7:  $\mathcal{T} \leftarrow$  top- $N_t$  maturities by volume; build  $\mathcal{K}_\tau$  ( $n_k$  per wing);  
pivot to  $\{\hat{\sigma}_{k,\tau}\}$ , cells  $\mathcal{N}$
- 8: **if** coverage (21) fails **then**
- 9:   clear stale diagnostics; **return**  $\emptyset$
- 10: **end if**
- 11:  $\hat{v} \leftarrow \text{clip}(\text{median}(\hat{\sigma})^2, [10^{-3}, 0.25])$ ; build seeds  $\mathcal{S}$  (28)
- 12: **for**  $s \in \mathcal{S}$  **do**
- 13:    $\hat{\Phi}_s \leftarrow$  LM minimiser of  $\sum_n (\varepsilon_n^{\text{px}})^2$  on  $\mathcal{B}$  from  $s$  (29)
- 14:   compute  $J_{\text{iv}}(\hat{\Phi}_s)$  (26)
- 15: **end for**
- 16:  $\Phi^* \leftarrow \arg \min_s J_{\text{iv}}(\hat{\Phi}_s)$  (select (30))
- 17: **if**  $J_{\text{iv}}(\Phi^*) \leq \varepsilon_{\text{iv}}$  **and**  $\neg \text{PEGGED}(\Phi^*)$  **then**
- 18:   write repriced diagnostics; **return**  $\text{row}(d, S_{\text{ref}}, r, g, \Phi^*, \mathcal{F}, J_{\text{iv}}, J_{\text{px}}, \dots)$
- 19: **else**
- 20:   clear stale diagnostics; **return**  $\emptyset$  (rejected)
- 21: **end if**

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